

**FIT NOURISH AI' MOBILE APP FOR
PROGRESS TRACKER & PREDICTIVE HEALTH
ALERT SYSTEM**

Project ID –25-26J-386

Project Proposal Report

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BSc (Hons) in Information Technology Specialization in Information Technology

Department of Information Technology
Sri Lanka Institute of Information Technology

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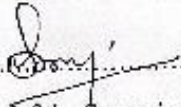
August 2025

Declaration

I declare that this is my own work, and this proposal does not incorporate without acknowledgement any material previously submitted for a degree or diploma in any other university or Institute of higher learning and to the best of our knowledge and belief it does not contain any material previously published or written by another person except where the acknowledgement is made in the text.

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The above candidate is carrying out research for the undergraduate Dissertation under my supervision.


Signature of the Supervisor

(Dr. Anjali Gamage)

29/8/2025
Date

Abstract

The rapid growth of wearable health devices and mobile applications has resulted in an abundance of real time, user generated health data, including nutrition records, calorie intake, weight, Body Mass Index (BMI), and physical activity logs. While these data streams hold significant potential for predictive health analytics, existing solutions often analyze such metrics in isolation or over short durations, limiting their ability to provide comprehensive and actionable insights.

This research proposes a neural network based predictive system capable of processing integrated, multi domain time series health data to forecast weight trends, detect nutrient imbalances, and identify risky behaviors such as excessive sodium consumption or early signs of fatigue. The proposed model leverages advanced deep learning architectures, including recurrent neural networks (RNNs), Long Short Term Memory (LSTM), and Gated Recurrent Units (GRU), to capture temporal dependencies and complex interactions between nutrition, activity, and physiological indicators.

By incorporating explainable AI techniques, the system aims to improve interpretability and foster user trust, making it more suitable for integration into real world health management platforms. The outcome of this research is expected to enable personalized, proactive wellness interventions, reduce long term health risks, and contribute to the advancement of AI driven preventive healthcare.

Keywords

Neural Networks, Time Series Prediction, Nutrition Analytics, Weight Forecasting, Personalized Health Monitoring, Risk Behavior Detection, Deep Learning, Explainable AI, Wellness Management

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List of Abbreviations

AI	Artificial Intelligence
ANN	Artificial Neural Network
API	Application Programming Interface
BMI	Body Mass Index
CNN	Convolutional Neural Network
DL	Deep Learning
GRU	Gated Recurrent Unit
IoT	Internet of Things
LSTM	Long Short-Term Memory
MAE	Mean Absolute Error
ML	Machine Learning
MUI	Material User Interface
NHANES	National Health and Nutrition Examination Survey
R^2	Coefficient of Determination
RNN	Recurrent Neural Network
RMSE	Root Mean Square Error
SHAP	SHapley Additive exPlanations
UI	User Interface
XAI	Explainable Artificial Intelligence

1.0 Introduction

The growing adoption of wearable health devices and mobile applications has led to an unprecedented availability of user generated health data, including detailed records of nutrition, calorie intake, weight, Body Mass Index (BMI), and physical activity. These continuous time series datasets present significant opportunities for advanced analysis and prediction in the domain of personalized health management.

Recent advances in deep learning, particularly neural network architectures, have demonstrated exceptional capability in modeling complex, non linear, and temporal dependencies in such multi dimensional health data. By leveraging these capabilities, it becomes possible to forecast weight trends, detect nutrient imbalances, and identify risky behaviors such as excessive sodium consumption or early signs of fatigue. These predictive insights can be integrated into proactive wellness management systems, enabling early interventions and tailored recommendations to improve health outcomes.

Unlike conventional statistical models, deep neural networks can automatically learn intricate patterns and correlations from large-scale, heterogeneous datasets without the need for extensive manual feature engineering. This advantage positions them as powerful tools for AI driven health analytics and predictive modeling. Integrating such models into user facing platforms has the potential to transform preventive healthcare from reactive responses to continuous, personalized monitoring and early risk detection.

This research focuses on developing a neural network based predictive system that utilizes time series health data to anticipate weight fluctuations, identify nutrition related health risks, and detect behavioral patterns indicative of potential health decline. The proposed solution aligns with the AI and Data Science specialization, aiming to bridge the gap between advanced machine learning methodologies and real world health monitoring applications.

1.1 Literature Review

Numerous studies have explored the application of neural networks in health prediction, emphasizing their effectiveness in modeling complex relationships between nutrition, physical activity, and physiological outcomes.

A multilayer perceptron (MLP) model was developed to classify weight status by integrating behavioral and nutritional inputs, achieving promising accuracy [1]. Similarly, a study conducted among healthcare professionals in North China demonstrated that deep neural networks could predict diet quality from BMI, lifestyle, and nutrition knowledge, highlighting the potential of AI in comprehensive dietary assessment [2].

Neural network ensembles have also been employed for body composition analysis, where multifactorial inputs were combined to compute fitness scores non invasively. Ensemble approaches demonstrated improved robustness in estimating complex health parameters compared to single model architectures [3].

During the COVID-19 pandemic, the need for temporal modeling became evident in a study that used deep neural networks to predict BMI fluctuations resulting from dietary and activity changes. The findings underscored the importance of incorporating longitudinal data to capture dynamic health trends [4].

In obesity risk prediction, comparative analyses of machine learning models have shown that integrating multi domain data covering physical activity, dietary habits, and environmental factors can significantly enhance prediction accuracy [5]. Likewise, the advantages of recurrent neural network architectures such as Long Short Term Memory (LSTM) and Gated Recurrent Units (GRU) in capturing sequential dependencies in healthcare time-series data have been emphasized [6].

The field of precision nutrition increasingly relies on machine learning to integrate diverse data types including food intake logs, biometric measurements, and metabolic responses to generate individualized dietary recommendations. A comprehensive review illustrated these advancements in nutrition research [7]. Transfer learning techniques have further advanced this area, showing how pre-trained deep recurrent networks on large-scale clinical datasets can be adapted for smaller, specialized datasets, thereby improving robustness and generalizability [8].

Furthermore, machine learning driven nutritional assessments have been successfully used to predict adverse clinical outcomes in vulnerable populations. Neural networks integrating nutrition and clinical data have been shown to effectively predict adverse events in older inpatients [9]. Reviews of machine learning applications in obesity prevention also reinforce the importance of

combining environmental, behavioral, and dietary data for timely risk detection and targeted preventive strategies [10].

Overall, the reviewed literature highlights significant progress in the use of neural networks for predicting health outcomes. However, most approaches are still limited in integrating multi-domain, longitudinal datasets for comprehensive and personalized predictions, underscoring the need for further research in this direction.

1.2 Research Gap

While existing research has made considerable progress in applying neural networks for health prediction, several critical gaps remain:

Lack of Integrated Multi Domain Time Series Modeling

Most current models analyze nutrition, physical activity, or clinical outcomes in isolation, rather than jointly modeling them as integrated, longitudinal datasets. This separation limits the comprehensiveness of health predictions.

Overreliance on Static or Short-Term Data

Many studies rely on cross-sectional or short-duration datasets, missing the value of continuous, long-term monitoring for early risk detection and trend analysis.

Limited Model Explainability

The interpretability of neural network predictions in personalized health contexts remains underexplored, reducing clinical acceptance and user trust.

Addressing these gaps requires developing an integrated neural network architecture capable of processing multi domain time series health data, incorporating interpretable AI methods, and leveraging diverse, representative datasets to produce accurate, personalized, and actionable health outcome prediction

Feature	Existing Research	My Research
Integrated Multi-Domain Time Series Data	✗	✓
Longitudinal Continuous Monitoring	✗	✓
Explainable / Interpretable AI	✗	✓

Table 1.2.1 Comparative Summary: Existing research and my Research

Dimension	Existing Solutions (Typical)	Proposed System (Neural Network + Time-Series + Explainability)	Novelty / Advantage
Scope	Focus on single domain (nutrition or activity or weight)	Multi-domain integration (nutrition + calories + activity + BMI + weight)	End-to-end holistic prediction
Temporal Modeling	Static snapshots or short-term models	Continuous time-series modeling using LSTMs/GRUs	Captures long-term health trends
Risk Detection	Limited to obesity/weight categories	Detects nutrient imbalance (e.g., sodium), fatigue risk, early warning	Proactive health risk alerts
Explainability	“Black-box” predictions, limited user trust	Integrated explainability (SHAP/LIME feature attribution)	Improves interpretability & trust
Adaptivity	Pre-trained or rule-based, requires manual updates	Learns continuously from new user data	Adaptive and personalized
Dataset Generalizability	Often trained on narrow demographic datasets	Uses transfer learning + diverse longitudinal data	Improves robustness & generalizability
User Engagement	Focus on backend prediction only	Interactive visual dashboard + personalized recommendations	Increases usability & adoption

Preventive Strategy	Reactive (after adverse event)	Preventive (early detection + proactive lifestyle suggestions)	Supports proactive wellness management
Security & Privacy	Basic storage (not always encrypted)	Secure data storage + encryption + access control	Enhances user data protection
Clinical Utility	Limited acceptance due to lack of transparency	Explainable, multi-feature insights for healthcare professionals	Better clinical trust & integration

Table 1.2. 2 Comparative Summary: existing solution and my solution

1.3 Research Problem

With the rapid growth of wearable health technologies and mobile applications, individuals are generating large volumes of real time health data, including nutrition logs, calorie intake, weight records, BMI measurements, and physical activity statistics. While these datasets present valuable opportunities for predictive analytics and proactive health management, current health monitoring systems largely rely on static assessments, isolated measurements, or short-term tracking.

This fragmented approach limits the ability to capture long term trends, identify early signs of health risks, or provide timely behavioral interventions. In particular, risky health behaviors such as excessive sodium intake, inadequate nutrient balance, or emerging fatigue are often detected only after noticeable health deterioration has occurred.

Although neural networks and other machine learning techniques have been applied in health prediction, most existing implementations focus on specific outcome variables rather than integrating multi domain time series data for comprehensive risk assessment. Furthermore, the interpretability of predictions remains a challenge, reducing trust among both clinicians and end users.

In the absence of integrated, interpretable, and longitudinal health prediction models, individuals miss the opportunity for early, data driven interventions that could prevent or mitigate chronic health conditions. Therefore, there is a pressing need for a robust, AI driven system capable of leveraging diverse time series health data to forecast weight trends, detect nutrient imbalances, and provide early alerts for risky behaviors.

2.0 Objectives

2.1 Main Objective

To develop a neural network based predictive system that utilizes integrated time series health data including nutrition, calorie intake, weight, BMI, and physical activity to forecast weight trends, detect nutrient imbalances, and identify early indicators of risky health behaviors, enabling proactive and personalized wellness management.

2.2 Specific Objectives

The following sub objectives will guide the research:

- Design and develop a deep learning model capable of processing multi-domain time-series health data for predictive analysis.
- Integrate nutrition, physical activity, and physiological metrics into a unified predictive framework for comprehensive health risk assessment.
- Implement early-warning mechanisms for detecting risky behaviors such as excessive sodium intake and fatigue risk.
- Enhance model interpretability using explainable AI techniques to increase clinical and user trust.
- Validate the predictive model using real world datasets to ensure robustness and generalizability across diverse demographic profiles.

3.0 Methodology

3.1 Requirement Gathering

Requirement gathering for this research was carried out through:

- Extensive review of prior research studies in the domains of nutrition analytics, time series health data modeling, and neural network based health prediction.
- Identification and analysis of existing mobile health and wearable based monitoring systems.
- Consultation with domain experts in nutrition and physical activity tracking to understand practical requirements.
- Exploration of public datasets such as NHANES and open health activity records to identify suitable data sources and formats.

3.1.1 Past Research Analysis

A wide range of research papers on weight prediction, nutrition monitoring, and time series deep learning models were reviewed. Key focus areas included:

- Neural network architectures for multi variable health prediction.
- Techniques for integrating nutrition and activity data into a unified prediction model.
- Time series modeling approaches such as LSTM, GRU, and hybrid models.

This analysis provided valuable insights into common challenges faced by researchers, such as data scarcity, overfitting in small datasets, and limited model explainability.

3.1.2 Identifying Existing Systems

Existing health monitoring applications such as MyFitnessPal, Apple Health, and Fitbit were studied to identify gaps in their predictive capabilities. While these platforms track user metrics, they often lack:

- Integration of nutrition, BMI, and activity data in a single predictive framework.
- Advanced AI-driven forecasting of weight and nutrient trends.
- Early warning systems for risky behaviors.

3.2 Feasibility Study

3.2.1 Technical Feasibility

The proposed solution requires knowledge and application of the following technologies:

- Recurrent Neural Networks (RNNs) – for capturing sequential dependencies in time-series data.
- Long Short Term Memory (LSTM) & GRU – for effective modeling of long-term dependencies.
- Convolutional Neural Networks (CNNs) – for feature extraction from certain structured or visualized data formats.
- Transfer Learning - to adapt pre-trained models from larger datasets for smaller, real-world health datasets.
- Explainable AI (XAI) methods – such as SHAP and LIME for enhancing interpretability.

3.2.1.2 Data Collection Knowledge

Data collection and preprocessing will involve:

- Data Acquisition – from publicly available health datasets and potential user trials.
- Data Cleaning – handling missing values, normalizing units, and removing anomalies.
- Feature Engineering – generating derived features such as weekly nutrient averages or activity trends.

3.2.2 Schedule Feasibility

The proposed research is planned for completion within the allocated academic project timeline, with separate phases for model development, testing, and integration.

3.2.3 Economic Feasibility

Since the solution is primarily software-based, the main cost components will be:

- Cloud-based computing resources for model training.
- Data storage and API integration costs.
- Minimal additional expenses for hosting and deployment.

3.3 System Analysis

3.3.1 Software Solution Approach

The development process will follow these key steps:

- Data Collection – acquiring multi-domain time-series health data.
- Data Preprocessing – cleaning, normalization, and handling missing entries.
- Feature Extraction – identifying relevant input variables for prediction.
- Model Selection & Training – testing different architectures (LSTM, GRU, hybrid models).
- Evaluation – using metrics such as MAE, RMSE, and R^2 for regression accuracy, and F1-score for classification tasks.
- Interpretability Analysis – applying explainable AI methods to improve model transparency.
- Integration – embedding the model into a user interface for real-time monitoring and predictions.

3.3.2 Tools & Technology

Core Programming Language

Python: The primary language for the entire backend and data science stack. It is the industry standard for machine learning, data analysis, and scientific computing due to its extensive ecosystem of libraries, simplicity, and strong community support.

1. Backend & API Development

This layer handles business logic, user authentication, database interactions, and serving data to the frontend.

Frameworks:

- Django: A high-level, "batteries-included" Python web framework. Excellent for rapid development with built-in admin panels, ORM (Object-Relational Mapper), and security features. Ideal if you need a monolithic structure with many built-in tools.
- Flask: A lightweight, micro-framework for Python. Offers more flexibility and is perfect for building RESTful APIs. You would choose Flask if you prefer a more modular approach, selecting your own libraries for each component (e.g., authentication, database ORM).

- **FastAPI:** A modern, high-performance web framework for building APIs with Python. Its key advantages are automatic interactive API documentation (Swagger UI), built-in data validation using Python type hints, and being asynchronous, which is crucial for handling numerous concurrent requests from wearables and user devices.

API Protocols:

- **REST:** Standard architectural style for stateless communication.
- **GraphQL:** An alternative to REST that allows the client to request exactly the data it needs, which can be more efficient for complex digital twin queries where different parts of the frontend need different data slices.

2. Machine Learning & Data Science Layer

This is the core of the personalization engine, handling prediction, optimization, and explainability.

Core Libraries:

- **Scikit-learn:** The fundamental library for traditional machine learning algorithms (linear regression, SVMs, clustering) and essential utilities (data preprocessing, model evaluation, feature selection).
- **XGBoost / LightGBM / CatBoost:** Premier libraries for building high-performance, gradient-boosted tree models. They are exceptionally effective for tabular data (which your user profiles and biometrics are) and provide built-in feature importance metrics for explainability.

Deep Learning (Optional for advanced biometric analysis):

- **TensorFlow / PyTorch:** Used if you plan to implement complex deep learning models, e.g., for analyzing time-series biometric data (sleep patterns, heart rate variability) with Recurrent Neural Networks (RNNs) or LSTMs.

Optimization:

- SciPy: For scientific computing and optimization algorithms. Its `scipy.optimize` module can be used for macro- and micronutrient constraint optimization.
- PuLP / CVXPY: Specialized libraries for linear programming (LP) and constraint optimization, which could be used by the "Planner & Optimization" module to generate meal plans that meet all nutritional and rule-based constraints.

Explainable AI (XAI):

- SHAP (SHapley Additive exPlanations): A game-theoretic approach to explain the output of any machine learning model. Crucial for your "Explainability & Audit" module to show users/doctors why a specific calorie target or macro split was recommended.
- LIME (Local Interpretable Model-agnostic Explanations): Another popular library for explaining individual predictions.

Data Processing & Manipulation:

- Pandas: The indispensable library for data manipulation and analysis. Used for cleaning, transforming, and aggregating all user, biometric, and lab data.
- NumPy: The foundation for numerical computation in Python, used by nearly all other data science libraries under the hood.

3. Data Storage & Databases

Different types of data require different storage solutions.

Primary Database (User Profiles, Static Data):

- PostgreSQL: A powerful, open-source object-relational database system. It's highly reliable, ACID-compliant, and has strong support for JSON data, which is useful for storing flexible biometric payloads.

Time-Series Database (Biometric Data from Wearables):

- InfluxDB / TimescaleDB: Specialized databases optimized for storing and querying time-stamped data (e.g., heart rate readings every 5 minutes). They offer superior performance for time-based queries and data downsampling compared to traditional SQL databases.

Graph Database (Digital Twin Relationships):

- Neo4j: A leading graph database. Perfect for your "BIM-Inspired Digital Twin" as it excels at storing and querying highly connected data. You can represent users, conditions, biometric readings, and food items as nodes, and the relationships between them (e.g., USER_HAS_CONDITION, RECOMMENDATION_AFFECTS_BIOMETRIC) as edges. This allows for powerful "what-if" traversals and simulations.

Caching (For Performance):

- Redis: An in-memory data store used as a cache to reduce latency for frequently accessed data (e.g., a user's current targets, frequently used rules) and for managing background job queues.

4. Frontend & Mobile

The user and clinician interface.

Web Framework:

- React.js: A dominant JavaScript library for building dynamic and responsive user interfaces. Its component-based architecture is ideal for creating reusable UI elements like dashboards, charts, and forms.

Mobile Development:

- React Native: Allows you to build native iOS and Android apps using React and JavaScript. This is an efficient choice as it lets you share a significant amount of code between your web and mobile applications.
- Native (Kotlin / Swift): For maximum performance and platform-specific integration, you would develop separate apps using Kotlin for Android and Swift for iOS.

Visualization:

- D3.js: A powerful low-level library for creating bespoke, interactive data visualizations (e.g., complex trend graphs for the digital twin simulation).

- Plotly / Recharts: Higher-level charting libraries that are easier to implement for standard charts (line, bar, pie) and integrate well with React.

5. Cloud, DevOps & Deployment

- Cloud Providers: AWS, Google Cloud Platform (GCP), or Microsoft Azure. They provide all the necessary scalable services: virtual machines (EC2/Compute Engine), managed databases (RDS, Cloud SQL), serverless functions (Lambda/Cloud Functions), and container orchestration (Kubernetes Engine/EKS).
- Containerization: Docker. To package the application and its dependencies into a standardized unit, ensuring consistency across different environments.
- Orchestration: Kubernetes (K8s). For automating the deployment, scaling, and management of containerized applications (your MDANE microservices).
- CI/CD: Jenkins, GitLab CI/CD, or GitHub Actions. To automate the testing and deployment process.

6. Third-Party Integrations & APIs

Wearable Device Integration: You would use the official APIs provided by:

- Apple HealthKit (for iOS devices)
- Google Fit (for Android devices)
- Fitbit Web API
- Garmin Health API

These APIs allow you to request user permission to read their health data (steps, heart rate, sleep) into your system.

3.4 Project Requirements

The success of the proposed health prediction system depends on clearly defined requirements that guide the design, development, and evaluation process. The requirements are categorized into functional, non-functional, user, and system requirements as outlined below.

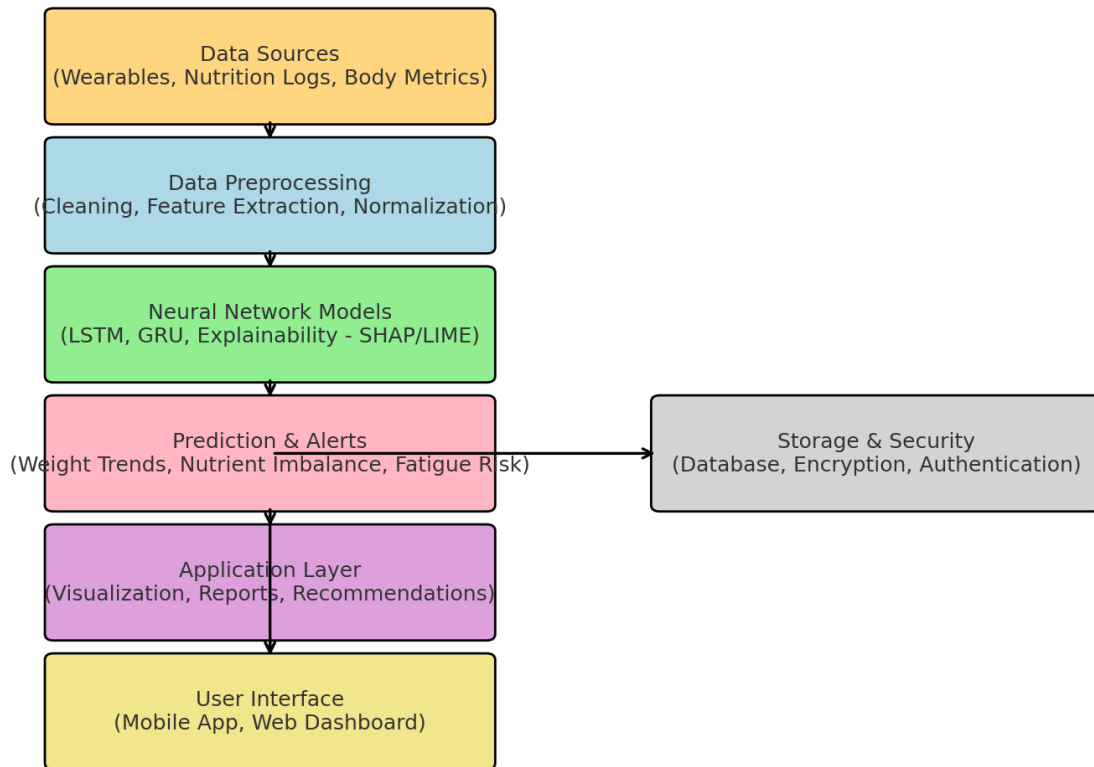


Figure 3.4.1 System Architecture diagram

3.4.1 Functional Requirements

The system must provide the following functionalities:

Data Acquisition

- Collect time-series data on nutrition, calorie intake, weight, BMI, and physical activity.
- Support both manual data entry (e.g., food logs) and integration with wearable/IoT devices.

Data Preprocessing & Management

- Handle missing, noisy, and inconsistent data.
- Normalize and store user health records securely in the database.

Health Outcome Prediction

- Forecast weight trends using neural network–based models.
- Predict nutrient imbalances (e.g., low protein, high sodium intake).
- Detect early signs of risky behaviors such as fatigue risk.

Risk Alert & Recommendations

- Generate personalized alerts when health risks are detected.
- Provide preventive lifestyle or nutrition recommendations based on prediction results.

Visualization & Reporting

- Display predictions and health trends in user-friendly dashboards.
- Generate summary reports for long-term monitoring.

Model Training & Updating

- Support retraining of the prediction model as new data becomes available.
- Allow transfer learning to adapt pre-trained models to new user populations.

3.4.2 Non-Functional Requirements

Usability

- Provide an intuitive, responsive, and visually clear interface suitable for non-technical users.

Reliability

- Ensure high prediction accuracy with minimal downtime.
- Provide consistent results across different devices and environments.

Performance

- Predictions should be generated in near real-time (within seconds).

- Efficient handling of large-scale time-series data without significant lag.

Security & Privacy

- Encrypt sensitive health data both in storage and during transmission.
- Ensure compliance with healthcare data protection standards (e.g., GDPR, HIPAA-like principles).

Scalability

- Support increasing numbers of users without performance degradation.
- Enable future integration with additional health metrics (e.g., sleep data, heart rate).

Interpretability

- Incorporate explainable AI methods to make predictions understandable to users and healthcare professionals.

3.4.3 User Requirements

End Users (General Public/Individuals)

- Ability to log nutrition and physical activity data manually.
- Receive daily/weekly predictions and risk alerts.
- Access personalized recommendations in an easy-to-understand format.

Healthcare Professionals

- Ability to review summarized patient data and predictive insights.
- Access interpretability features to understand the rationale behind predictions.
- Download or share patient reports for clinical decision-making.

System Administrators

- Manage user accounts and permissions.
- Monitor system performance and data integrity.

- Update and maintain predictive models.

3.4.4 System Requirements

Hardware Requirements

- Server/Cloud infrastructure for model training and deployment.
- GPU-enabled machines for deep learning model training.
- User devices: smartphones, tablets, or PCs for accessing the system.

Software Requirements

- Backend: Python, TensorFlow/Keras, Scikit-learn for model training.
- Frontend: React.js with Material UI for interactive dashboards.
- Database: MongoDB for flexible storage of time-series health data.
- Deployment: Docker and Kubernetes for scalable deployment.

Data Requirements

- Access to publicly available datasets (e.g., NHANES, open nutrition datasets).
- Collection of real-world user data for model testing and refinement.

Integration Requirements

- APIs to connect with wearable devices (e.g., Fitbit, Apple Health, Google Fit).
- Compatibility with third-party health platforms for extended functionality.

3.5 Testing

Testing plays a critical role in ensuring the accuracy, reliability, and usability of the proposed health prediction system. Since the project involves both a machine learning model and a user-facing application, the testing process will be carried out in multiple phases.

3.5.1 Model Testing

The predictive neural network will be tested using:

- **Train/Test Split & Cross-Validation** – To evaluate model performance on unseen data and reduce overfitting.
- **Evaluation Metrics** – Mean Absolute Error (MAE), Root Mean Square Error (RMSE), and R^2 for regression tasks such as weight forecasting; Precision, Recall, and F1-score for classification tasks such as risky behavior detection.
- **Benchmarking** – Model results will be compared against baseline statistical methods (e.g., linear regression) and alternative machine learning approaches (e.g., Random Forest, Support Vector Machines).
- **Robustness Testing** – The model will be tested on noisy and incomplete datasets to ensure resilience in real-world conditions.

3.5.2 System Testing

The overall system will be validated through:

- **Unit Testing** – Each module (data preprocessing, prediction, visualization) will be tested individually for correctness.
- **Integration Testing** – Interactions between components such as the neural network, database, and user interface will be tested to ensure smooth data flow.
- **System Testing** – The end-to-end system will be evaluated in a controlled environment to verify full functionality.

3.5.3 User Testing

To ensure usability and effectiveness:

- **Alpha Testing** – Conducted internally among the research team to identify functional errors and interface issues.
- **Beta Testing** – Conducted with a selected group of real users (e.g., volunteers tracking their nutrition and activity data) to evaluate system usability, clarity of predictions, and relevance of recommendations.

- **User Feedback Surveys** – Structured questionnaires and interviews will collect user feedback on system accuracy, ease of use, and trust in the recommendations.

3.5.4 Acceptance Testing

The final phase will focus on ensuring that the proposed system meets the research objectives and requirements:

- **Functional Acceptance** – Verifying that all functional requirements (prediction, alerts, visualization) are met.
- **Non-Functional Acceptance** – Evaluating usability, performance, and security compliance.
- **Stakeholder Validation** – Engaging supervisors, domain experts, and selected end-users to confirm the practical value of the system.

3.6 Timeline

The proposed project will be completed within the allocated academic year, with work divided into distinct phases. Each phase has defined deliverables to ensure systematic progress.

Proposed Timeline for Project Activities

Task	Start Date	End Date	Status
Brainstorming workshop	20/05/2025	25/05/2025	Done
Group registration	25/05/2025	30/05/2025	Done
Topic Assessment Form (TAF)	01/07/2025	15/07/2025	Done
Project charter	15/07/2025	15/08/2025	Done

Proposal report (draft)	15/08/2025	15/09/2025	Progress
Proposal presentation	15/09/2025	20/09/2025	Pending
Proposal report (final)	20/09/2025	15/12/2025	Pending
Check list 1	15/12/2025	20/12/2025	Pending
Research paper	20/12/2025	30/03/2026	Pending
Final report	30/03/2026	30/04/2026	Pending
Progress presentation 2	15/08/2025	30/03/2026	Pending
Check list 2	15/08/2025	30/03/2026	Pending
Submission of RP			
Final presentation and viva	15/08/2025	30/03/2026	Pending
Project application			
Research logbook	15/08/2025	30/06/2026	Pending

Table 3.6.1 Proposed Timeline for Project Activities

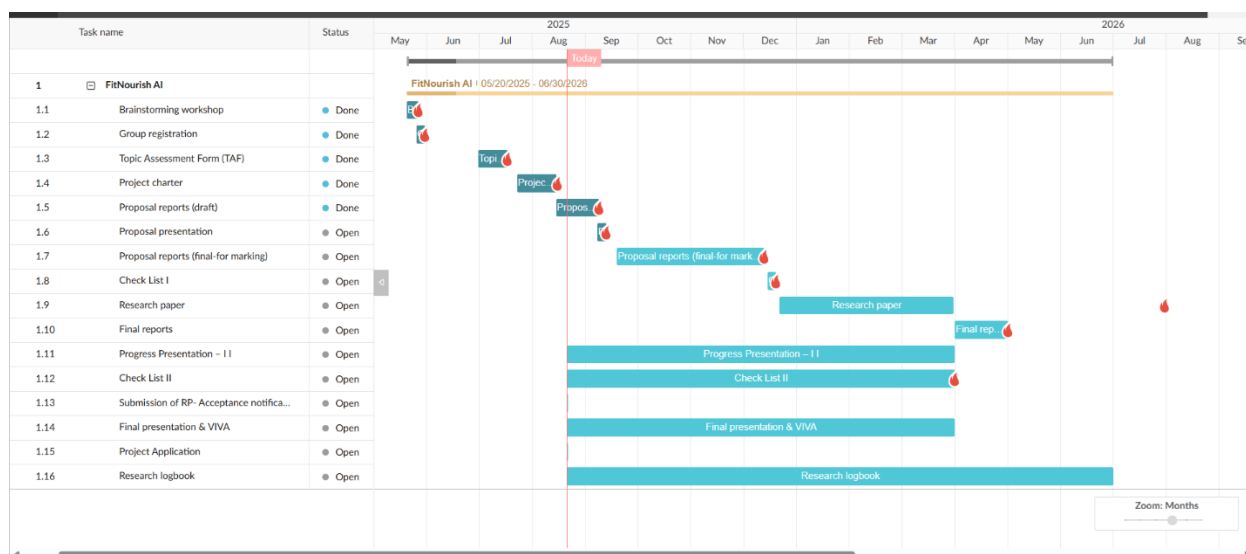


Figure 3.6.2 Gantt chart

3.7 Risk Management Plan

Risk management is essential to ensure the smooth execution of the research project. The following table outlines the potential risks, their triggers, responsible parties, and mitigation strategies.

Risk	Trigger	Owner	Response / Mitigation Strategy	Resources Required
Data Availability Issues	Lack of access to sufficient real-world or public datasets	Project Team	Use multiple open datasets (e.g., NHANES, Kaggle); apply data augmentation techniques; simulate synthetic data if needed	Public datasets, data preprocessing tools
Data Privacy & Security Concerns	Handling sensitive health-related data	Project Team & Supervisor	Anonymize datasets, apply encryption, and comply with data protection standards	Secure servers, encryption libraries
Model Overfitting / Poor Generalization	Limited or imbalanced dataset during training	ML Developer	Apply cross-validation, regularization, and dropout; use transfer learning on pre-trained models	TensorFlow/Keras, GPU resources
System Performance Bottlenecks	Large-scale time-series data slows	Backend Developer	Optimize database queries, use batch processing, and	MongoDB Atlas, cloud GPU/CPU instances

	down predictions		scale using cloud resources	
Interpretability Challenges	Neural network predictions are difficult to explain to users	ML Developer & Research Team	Incorporate Explainable AI (XAI) methods (SHAP, LIME); provide simplified explanations in UI	XAI libraries, visualization tools
User Adoption Issues	Users may not trust or engage with predictions	Research Team & Frontend Developer	Conduct user testing; improve UI design for simplicity; provide transparent explanation of results	User feedback surveys, UI design tools
Time Constraints	Delays in development or integration phases	Project Team Leader	Strict project management; regular progress reviews; reallocate resources if needed	Gantt chart, project management tools
Hardware/Software Failures	Server downtime or tool incompatibility	System Administrator	Maintain backups; use containerization (Docker); deploy on reliable cloud platforms	Cloud service, Docker/Kubernetes

Table 3.7.1 Risk Management Plan

3.8 Communication Management Plan

Effective communication is crucial for the successful execution of the proposed research project. The communication plan ensures that all stakeholders including the research team, supervisor, and co-supervisor receive timely and accurate information regarding progress, risks, and outcomes.

3.8.1 Communication Objectives

- Ensure proactive and transparent communication among all stakeholders.
- Provide progress updates regularly to supervisors and team members.
- Share technical challenges, risks, and decisions promptly.
- Maintain proper documentation of discussions, decisions, and milestones.
- Facilitate efficient collaboration and task management.

3.8.2 Communication Media

1. The following communication channels will be used throughout the project:
2. Email – For formal communication and documentation sharing.
3. Meetings – Weekly/bi-weekly discussions with supervisor and co-supervisor via MS Teams/Zoom.
4. Messaging Platforms – WhatsApp/Slack for daily quick updates and informal coordination.
5. Project Management Tools – MS Planner or Trello for tracking tasks and deadlines.
6. Documentation – Shared Google Docs/Drive for collaborative writing and record-keeping.

3.8.3 Communication Plan Table

Meeting Type	Attendees	Purpose	Frequency	Agenda Items
Kick-off Meeting	Supervisor, Co-supervisor, Project Team	Formally initiate the project, define scope, expectations, and deliverables	Once at project start	Project scope, objectives, timeline, risk discussion
Requirement & Design Meetings	Supervisor, Co-supervisor, Project Team	Review requirements and system design	Weekly during design phase	Requirement validation, model selection, architecture
Progress Review Meetings	Supervisor, Co-supervisor, Project Team	Monitor project progress and identify challenges	Bi-weekly	Current progress, risks/issues, next milestones
Internal Team Meetings	Project Team	Discuss day-to-day tasks, assign responsibilities	Twice a week	Task updates, blockers, technical issues
Pre-Presentation Reviews	Supervisor, Co-supervisor, Project Team	Evaluate readiness for proposal, progress presentations, and final viva	Before major deliverables	Demo, presentation review, feedback incorporation
Final Review & Debrief	Supervisor, Co-supervisor, Project Team	Summarize achievements, lessons learned, and final outcomes	Once at project end	Project outcomes, challenges, best practices

Table 3.8.3.1 Communication Plan

3.9 Use Cases

The proposed system aims to support multiple stakeholders, including individual users, healthcare professionals, and wellness organizations. The following use cases illustrate how the system will be applied in practice:

Use Case 1: Weight Trend Prediction

Actors: Individual User

Description: A user logs daily calorie intake and activity data via the application or wearable integration. The system analyzes the time-series data and forecasts future weight trends.

Outcome: The user receives a personalized prediction of expected weight changes for the next 4–8 weeks, enabling proactive dietary adjustments.

Use Case 2: Nutrient Imbalance Detection

Actors: Individual User

Description: A user's nutrition logs reveal consistently high sodium and low protein intake. The system detects the imbalance using neural network analysis and issues a personalized alert.

Outcome: The user is notified about the imbalance and receives dietary recommendations to restore nutritional balance.

Use Case 3: Fatigue Risk Alert

Actors: Individual User

Description: Based on activity patterns and calorie intake, the system identifies early indicators of potential fatigue risk (e.g., low energy balance over several days).

Outcome: The user is alerted to take preventive actions, such as rest or improved nutrition, reducing the chance of exhaustion.

Use Case 4: Clinical Monitoring Support

Actors: Healthcare Professional

Description: A doctor or nutritionist accesses a summary report of a patient's health trends, nutrient intake, and risk alerts generated by the system.

Outcome: The professional uses the insights to provide more personalized treatment or lifestyle guidance, improving patient care quality.

Use Case 5: Wellness Program Integration

Actors: Corporate Wellness Team / Fitness Center

Description: An organization integrates the system into its wellness program to monitor employee or client health trends. Risk alerts (e.g., obesity, fatigue, poor nutrition) are tracked across participants.

Outcome: Early interventions (e.g., tailored diet plans, fitness schedules) are implemented, promoting better overall wellness and reducing healthcare costs.

3.10 Test cases

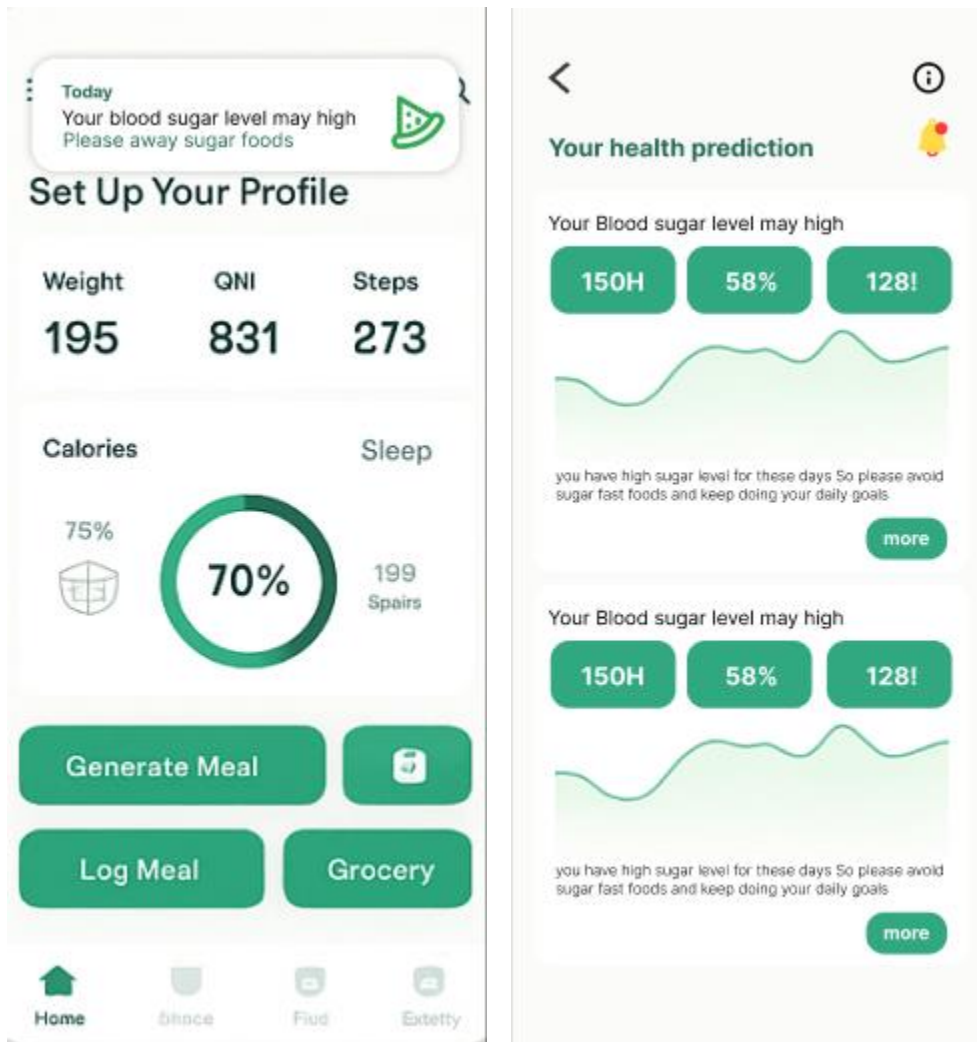
The following test cases are designed to validate the functionality, performance, and usability of the proposed system

Test Case ID	Test Scenario	Precondition	Test Steps	Expected Result
TC-01	Verify data entry (manual nutrition log)	User is logged into the system	1. Navigate to nutrition log section. 2. Enter meal details (calories, sodium, protein, etc.). 3. Save record.	System successfully saves nutrition log and updates database.
TC-02	Verify data import from wearable	Wearable device is connected	1. Connect wearable (e.g., Fitbit). 2. Sync activity data.	System imports steps, calories burned, and activity data correctly.
TC-03	Verify weight trend prediction	At least 30 days of user data available	1. Select “Weight Prediction” option. 2. Run forecast.	System displays predicted weight trend for next 4–8 weeks with graph.
TC-04	Detect nutrient imbalance	User has logged nutrition for at least 7 days	1. Analyze logged nutrition data. 2. Run imbalance check.	System generates alert if sodium > threshold or protein < threshold.

TC-05	Detect fatigue risk	Continuous activity + calorie deficit recorded	1. Analyze last 7 days of activity and intake. 2. Check fatigue risk model.	System triggers fatigue risk alert and provides recommendations.
TC-06	Verify model accuracy	Test dataset available	1. Run prediction model on test dataset. 2. Compare predictions vs. actual outcomes.	Accuracy meets threshold (e.g., RMSE < 5%, F1-score > 0.8).
TC-07	Verify explainability	Prediction generated	1. Request explanation for prediction.	System provides SHAP/LIME-based feature contribution explanation.
TC-08	Verify report generation	User has at least 1 month of data	1. Navigate to Reports section. 2. Generate monthly health report.	System generates PDF/visual report with trends and insights.
TC-09	Verify data security	User submits sensitive data	1. Submit/update health data. 2. Inspect storage & transfer logs.	Data is encrypted in database and during transmission.
TC-10	Verify usability	End-user testing session	1. Provide system to selected test users. 2. Collect feedback via survey.	Users report system as easy to use, predictions understandable, UI responsive.

Table 3.10.1 Test cases

3.11 UI design



4.0 Commercialization

The proposed health prediction system has strong potential for commercialization due to the growing global demand for AI-driven wellness and preventive healthcare solutions. With increasing adoption of wearable devices and health tracking applications, individuals are seeking personalized insights and proactive health management tools.

The system can be commercialized in the following ways:

1. Integration with Existing Health Platforms
 - The predictive engine can be integrated into popular applications such as Fitbit, Apple Health, and Google Fit to provide advanced forecasting and risk alerts beyond simple tracking.
2. Mobile Health Application
 - A standalone mobile application can be developed for end-users, offering weight trend predictions, nutrient imbalance alerts, and lifestyle recommendations.
 - The app can follow a freemium model: free basic features (logging, tracking, visualization) with premium features (personalized AI-driven recommendations, risk alerts).
3. Subscription Model for Wellness Organizations
 - Fitness centers, nutritionists, and wellness consultants can use the system as a subscription-based platform to monitor clients' health trends and deliver data-driven guidance.
4. Healthcare Partnerships
 - Hospitals and clinics can adopt the system to support preventive healthcare programs, chronic disease management (e.g., obesity, hypertension), and nutrition counseling.
5. Corporate Wellness Programs
 - Organizations can deploy the system to monitor employee wellness, reduce fatigue-related risks, and improve productivity through early interventions.

By commercializing the solution across these sectors, the system can contribute to reducing long-term healthcare costs, improving lifestyle management, and promoting preventive healthcare on a large scale.

4.0 Budget

Budget Estimation and Justification

A preliminary budget has been prepared to cover the essential resources required for the successful implementation and evaluation of the proposed system. Table 1 summarizes the anticipated costs.

Category	Description	Estimated Cost (LKR)
Hardware / Devices	Samsung Galaxy Watch 6 – for biometric data acquisition (heart rate, SpO ₂ , sleep, stress monitoring).	50000
Software / Tools	Python, TensorFlow, PyTorch, Google Colab (open source / free tiers).	0
Cloud / Hosting	GPU-enabled cloud resources (Google Colab Pro / AWS EC2) for model training.	6000
Datasets	Publicly available datasets (free) and premium access where necessary.	2000
Documentation & Printing	Report preparation, printing, and binding.	3000
Miscellaneous	Internet, electricity, and contingency costs.	4000
Total Estimated Budget		65000

Justification

The primary cost arises from the acquisition of the Samsung Galaxy Watch 6, which will be used to collect real-time biometric data including heart rate, SpO₂ levels, stress, and sleep quality. This device was selected due to its high accuracy of sensors, comprehensive health monitoring features, and SDK/API compatibility, which ensures seamless integration with the proposed nutrition and wellness system. Although lower cost alternatives such as fitness bands exist, their limited sensor reliability and restricted developer support make them less suitable for research validation. Other costs are minimal since most of the required software tools are open-source. However, limited expenditure is allocated for cloud-based GPU resources to accelerate deep learning model training, as well as for dataset acquisition, documentation, and contingency requirements. Overall, the total budget of LKR 65,000 is justified, with the smartwatch representing the critical hardware investment necessary to ensure data accuracy, system feasibility, and reliable evaluation outcomes.

5.0 Summary

This research proposes the development of a neural network–based predictive system that integrates multi-domain time-series health data — including nutrition, calorie intake, weight, BMI, and physical activity — to forecast health outcomes and detect early signs of risky behaviors. By leveraging advanced deep learning architectures such as LSTM and GRU, the system is designed to capture complex temporal dependencies, providing accurate and proactive wellness insights.

Unlike existing health tracking applications, this solution goes beyond data logging by offering predictive analytics, early-warning alerts, and interpretable recommendations. The inclusion of Explainable AI (XAI) ensures transparency, which is critical for building user trust and potential adoption in healthcare settings.

The outcome of this research is expected to:

- Enable personalized and preventive health management.
- Assist individuals in making better lifestyle and dietary decisions.
- Provide healthcare professionals with data-driven insights for improved patient care.
- Contribute to the growing field of AI in public health and precision nutrition.

In summary, this project aims to bridge the gap between raw health data and actionable intelligence, ultimately empowering individuals and organizations to move from reactive healthcare to proactive wellness management.

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Figure 5.1 Plagiarism checking